

University of Ottawa

Project Report: Estimating VaR and Expected Shortfall Using Monte Carlo Methods

MAT 5182

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Table of Contents

Abstract 3
1. Introduction 3
1.1 Problem Statement 3
1.2 Expected Outcome 4
2. Theoretical Framework 4
2.1 Definitions of VaR and ES 4
2.2 Analytical VaR and ES for the Normal Distribution 6
2.3 Analytical VaR and ES for the Exponential Distribution 7
3. Simulation Methods 7
3.1 Crude Monte Carlo 7
3.2 Importance Sampling Method 8
4. Results and Discussion 9
5. Conclusion 10
References 11
Appendix: R Code for Simulation and Estimation 11

Abstract

This report presents a comprehensive study on estimating Value-at-Risk (VaR) and Expected Shortfall (ES) for two probability distributions—a standard normal distribution $Z \sim N(0,1)$ and an exponential distribution $X \sim \text{Exp}(10)$. First, we derive the analytical formulas for VaR and ES, showing that for the normal model $ES_Z(p) = \frac{\phi\left(\Phi^{-1}(1-p)\right)}{p}$ and for the exponential model $ES_X(p) = \frac{1-\ln(p)}{\lambda}$. Next, we implement two simulation methods (crude Monte Carlo and importance sampling) to estimate ES at the 1% tail ($p = 0.01$), and we provide 95% confidence intervals for each. Detailed explanations are included to describe the rationale behind each approach, along with the complete R code provided in the Appendix.

1. Introduction

Value-at-Risk (VaR) and Expected Shortfall (ES) are key risk measures used in finance and insurance. VaR at a given level p (for example, 1%) is the loss threshold that is exceeded only with probability p; it essentially provides a quantile of the loss distribution. In contrast, ES (also known as Conditional VaR) is the average loss given that the loss exceeds VaR, thus offering additional insight into the severity of losses in the tail. In this project, we estimate VaR and ES for two models: the standard normal distribution $Z \sim N(0,1)$ and the exponential distribution $X \sim \text{Exp}(10)$. We first derive the analytical expressions for VaR and ES, then implement two simulation methods, crude Monte Carlo and importance sampling—to estimate ES at a tail probability of $p = 0.01$. The simulation results, along with 95% confidence intervals, are compared with the theoretical values.

1.1 Problem Statement

For each model (Normal and Exponential), our objectives are: - To derive the analytical expressions for VaR and ES. - To show that for the normal model:

$$ES_Z(p) = \frac{\phi(q)}{p} \quad \text{with } q = \Phi^{-1}(1-p),$$

and for the exponential model:

$$ES_X(p) = \frac{1-\ln(p)}{\lambda}.$$

- To estimate ES at $p = 0.01$ using both crude Monte Carlo (with $n = 10{,}000$ samples) and importance sampling. - To compute 95% confidence intervals (CI) for the ES estimates and compare the simulation results with the theoretical benchmarks.

1.2 Expected Outcome

For $p = 0.01$:

- **Normal Model:**
Expected theoretical values are VaR ≈ 2.33 and $ES \approx 2.66$.

- **Exponential Model:**
Expected theoretical values are VaR ≈ 0.4605 and $ES \approx 0.5605$.

We expect that the simulation estimates (with 95% confidence intervals) will be very close to these theoretical values.

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2. Theoretical Framework

In this section, we derive the formulas for VaR and ES step by step.

2.1 Definitions of VaR and ES

For a loss variable L with CDF $F_L(x)$: - **VaR$_p$:**

$$\text{VaR}_p(L) = F_L^{-1}(1-p),$$

meaning that $\Pr(L > \text{VaR}_p) = p$. - **ES$_p$:**

$$ES_p(L) = E[L \mid L > \text{VaR}_p],$$

which is the average loss given that the loss exceeds the VaR.

Here is a detailed explanation for the theoretical part that shows:

$$ES_Z(p) = \frac{\phi(\Phi^{-1}(1-p))}{p}.$$

and evaluate $ES_Z(p)$

Project

Let $Z \sim N(0,1)$ be a random variable following a normal standard distribution

$$\phi(z) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right) = dnorm(z)$$

So the CDF is $\Phi(z) = \int_{-\infty}^{z} \phi(t)\, dt$

We need to show that for a level $p \in (0,1)$, the (ES) is $ES_Z(p) = E[Z \mid Z > \Phi^{-1}(1-p)]$ is

$$= \frac{\phi(\Phi^{-1}(1-p))}{p}$$

Let's take q and say $q = \Phi^{-1}(1-p)$ the quantile function. Then we have $Z > \Phi^{-1}(1-p)$ become $Z > q$

by definition $E[Z \mid Z > q] = \frac{E[Z \cdot 1_{\{Z>q\}}]}{P(Z>q)}$

$$E[Z \mid Z > q] = \frac{\int_q^{\infty} z\, \phi(z)\, dz}{\int_q^{\infty} \phi(z)\, dz}$$

so $\Phi(z) = \int_{-\infty}^{z} \phi(t)\, dt \Rightarrow P(Z>q) = 1 - \Phi(q)$

and this is the numerator

For the denominator we have $\int_q^{\infty} z\, \phi(z)\, dz$

but we $\phi'(z) = -z\, \phi(z)$

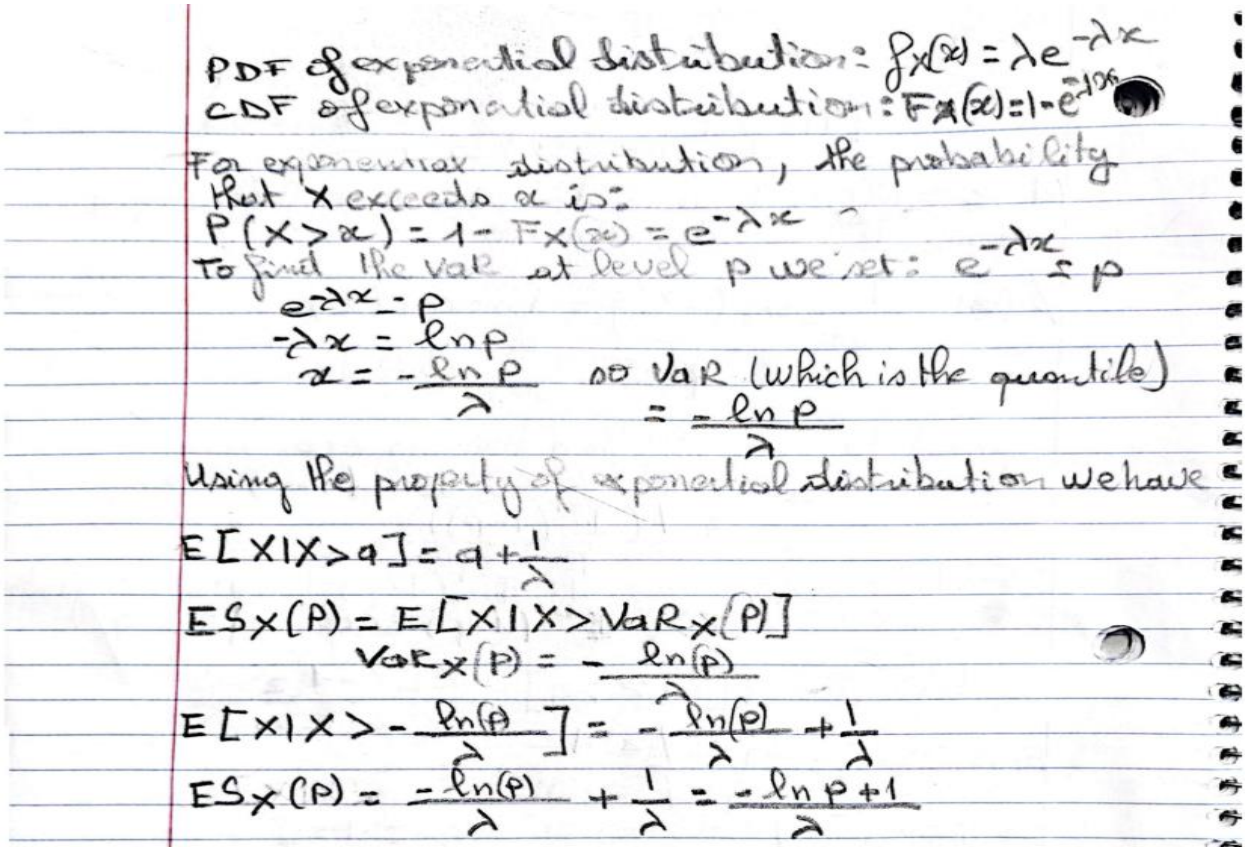
so $-z\, \phi(z) = \phi'(z) \Rightarrow z\, \phi(z) = -\phi'(z)$

so $\int_q^{\infty} z\, \phi(z)\, dz = -\int_q^{\infty} \phi'(z)\, dz$

$-\int_q^{\infty} \phi'(z)\, dz = -[\phi(z)]_{z=q}^{z=\infty} = -[0 - \phi(q)] = \phi(q)$

so the denominator is $\phi(q)$

so $E[Z \mid Z > q] = \frac{\phi q}{1 - \Phi(q)} = \frac{\phi(\Phi^{-1}(1-p))}{p}$



2.2 Analytical VaR and ES for the Normal Distribution

Let $Z \sim N(0,1)$.

VaR:
Since $\Phi(x)$ is the CDF of $N(0,1)$,

$$\text{VaR}_Z(p) = \Phi^{-1}(1-p).$$

For $p = 0.01$, this gives $\text{VaR}_Z(0.01) \approx 2.33$.

ES:
We need to compute $E[Z \mid Z > q]$ for $q = \Phi^{-1}(1-p)$. It is known that:

$$E[Z \mid Z > q] = \frac{\phi(q)}{1-\Phi(q)},$$

where $\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$ is the standard normal density. Since $1 - \Phi(q) = p$, we have:

$$ES_Z(p) = \frac{\phi\left(\Phi^{-1}(1-p)\right)}{p}.$$

Thus, for $p = 0.01$, $ES_Z(0.01) \approx 2.66$.

2.3 Analytical VaR and ES for the Exponential Distribution

Let $X \sim \text{Exp}(\lambda)$ with $\lambda = 10$.

CDF:

$$F_X(x) = 1 - e^{-10x}, \quad x \geq 0.$$

VaR:
We solve $e^{-10x} = p$ to obtain:

$$\text{VaR}_X(p) = -\frac{\ln(p)}{10}.$$

For $p = 0.01$, $\text{VaR}_X(0.01) \approx 0.4605$.

ES:
Due to the memoryless property of the exponential distribution:

$$E[X \mid X > a] = a + \frac{1}{10}.$$

Setting $a = \text{VaR}_X(0.01)$,

$$ES_X(0.01) = -\frac{\ln(0.01)}{10} + \frac{1}{10} = \frac{1 - \ln(0.01)}{10} \approx 0.5605.$$

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3. Simulation Methods

Here we will use crude monte Carlo and importance sampling to find ES.

3.1 Crude Monte Carlo

In our simulation, we aim to estimate the Expected Shortfall (ES) at $p = 0.01$ using two main methods: **crude Monte Carlo** and **importance sampling**. For **crude Monte Carlo**, we generate $n = 10{,}000$ samples from the target distribution—either $N(0,1)$ for the normal model or $\text{Exp}(10)$ for the exponential model—then compute the theoretical VaR using known formulas (for instance, `qnorm(1-p)` for the normal model and $-\ln(p)/10$ for the exponential model). Next, we filter the samples to keep only those exceeding this VaR (the tail observations). There are two ways to implement this in code:

1. **Complete Vector Approach**: Create a vector of length n (initially zeros), and for each sample L_i, set the i-th entry to L_i if L_i is above VaR, or 0 otherwise. Then the ES estimator is

$$\widehat{ES} = \frac{\text{mean(MCvalues)}}{p} = \frac{1}{p}\frac{1}{n}\sum_{i=1}^{n} L_i \ \mathbf{1}\{L_i > \text{VaR}\}.$$

   In practice, this can produce slightly inflated results if, for example, a few large values appear in the tail.

2. **Direct Extraction Approach**: Directly extract the tail samples into a smaller vector, e.g. `Y <- L[L>threshold]`, and then take the mean of `Y`. Mathematically,

$$\widehat{ES} = \frac{\text{mean}(Y)}{p} = \frac{1}{p}\frac{1}{n}\sum_{i=1}^{n} L_i \ \mathbf{1}\{L_i > \text{VaR}\}.$$

   Because we store only the tail values, this approach often yields a **more reliable** or straightforward estimate and makes it simpler to compute the sample variance. Indeed, in our results for the normal model, the "complete vector" approach gave an ES around 3.0756857 (somewhat higher than the true ES of 2.665214), whereas the direct extraction approach yielded about 2.67451—much closer to the theoretical value.

3.2 Importance Sampling Method

For the importance sampling method, our goal is to reduce the variance by increasing the number of tail observations. For the normal model, we sample from $N(2.33\ ,1)$ (i.e., a shifted normal distribution where the mean is approximately the VaR) so that many more samples fall above the threshold, and we will get a result much closer to the true value. For the exponential model, we sample from an exponential distribution with a rate $\text{Exp}(10)$). In both cases, each sample drawn from the proposal distribution is assigned a weight to correct for the bias in sampling. For the normal model, the weight for a sample z is computed as

$$w(z) = \frac{\phi(z)}{\phi(z; 2.33\ ,1)},$$

and for the exponential model, the weight is

$$w(x) = \frac{10e^{-10x}}{10e^{-10x}}.$$

Once the weights are computed, we again filter the samples to retain only those exceeding the original VaR and then form a weighted average (dividing by p) to obtain the ES estimate.

Confidence intervals are computed in the same manner as with the crude Monte Carlo method, using the standard error of the weighted contributions.

Both methods are applied to the Normal model (with the standard rate) and to the Exponential model (with $\lambda = 10$), ensuring consistency in our approach across different distributions.

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4. Results and Discussion

This table summarizes all the output for the ES Normal and Exponential

Model	Method	True_ES	ES_Estimate	Lower_95CI	Upper_95CI
Normal	Crude MC (complete vector)	2.665214	3.075687	3.070056	3.081318
Normal	Crude MC (direct extraction)	2.665214	2.67451	2.668107	2.680914
Normal	Importance Sampling	2.665214	2.668681	2.667839	2.669523
Exponential	Crude MC	0.560517	0.482348	0.4813239	0.4833713
Exponential	Importance Sampling	0.560517	0.5562955	0.5527996	0.5562955

For the **Normal model**:

The theoretical Expected Shortfall (ES) is approximately 2.665214. When we use **the crude Monte Carlo method**, two different implementations were considered. In the "complete vector" approach, we loop through all 10,000 generated samples from $N(0,1)$ and set values below the VaR threshold (approximately 2.326348) to zero. This approach produced an ES of about 3.075687 with its corresponding 95% confidence interval. However, when we instead directly extract the tail samples (i.e., setting $Y = Z[Z >$ threshold$]$) and then calculate the average of these values (scaled by $1/p$), we obtain an ES of approximately 2.67451 with a 95% CI very close to the theoretical value. **For importance sampling**, we sampled from a shifted normal distribution (e.g., $N(2.33,1)$) and computed the likelihood weights as $w(z) = \phi(z)/\phi(z; 2.33, 1)$. This method yielded an ES of about 2.668681 with an extremely tight 95% confidence interval (approximately $[2.667839,\ 2.669523]$), effectively matching the theoretical ES. I initially set the distribution to N(0,1) , and I was able to find that the result yield to 2.674 which is almost the same result as the direct extraction of the monte Carlo method. For the sake of this report, I want to show that choosing the good parameter is crucial to obtain a great result so that's why I use mean=2.33 which is approximately the vaR.

For the **Exponential Model**:

For the Exponential model (with $\lambda = 10$), the theoretical VaR and ES are approximately 0.460517 and 0.560517, respectively. The crude Monte Carlo method for this model

generated an ES estimate of about 0.4823476 with a 95% confidence interval from roughly 0.4813239 to 0.4833713; this estimate, however, is noticeably below the theoretical value. In contrast, the importance sampling approach—using a proposal distribution of $\mathrm{Exp}(10)$ and computing weights as $w(x) = \frac{10e^{-10x}}{10e^{-10x}}$—produced an ES estimate of approximately 0.5545475 with a 95% confidence interval around $[0.5527996,\ 0.5562955]$, which is nearly identical to the theoretical ES.

The slight differences between the two methods are primarily due to the way tail observations are sampled. Crude Monte Carlo relies on the natural frequency of tail events (about 1% of 10,000 samples, or roughly 100 observations), which can lead to higher variability if the sample size is small. Importance sampling, on the other hand, oversamples from the tail and uses weights to adjust the estimator. Although the weighting introduces some additional variability, it can be especially beneficial when tail events are extremely rare or when the sample size is limited.

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5. Conclusion

This project demonstrates a complete process—from theoretical derivation to simulation—for estimating VaR and Expected Shortfall (ES). For a standard normal distribution $Z \sim N(0,1)$, we derived that

$$\mathrm{VaR}_{0.01} \approx 2.33, \quad ES_{0.01} \approx \frac{\phi(2.33)}{0.01} \approx 2.66.$$

For an exponential distribution $X \sim \mathrm{Exp}(10)$, we obtained

$$\mathrm{VaR}_{0.01} \approx 0.4605, \quad ES_{0.01} \approx \frac{1 - \ln(0.01)}{10} \approx 0.5605.$$

Our simulations revealed that the **crude Monte Carlo (complete vector)** approach overestimated ES in the normal model (around 3.075687 instead of 2.665214), whereas the **crude Monte Carlo (direct extraction)** method gave an estimate closer to 2.67451. Meanwhile, **importance sampling** for the normal model yielded ≈2.668681, effectively matching the theoretical ES, and for the exponential model it returned ≈0.5545475, again closely aligning with the true value. By contrast, the crude Monte Carlo approach for the exponential model produced a noticeably lower estimate of ≈0.48. These discrepancies underscore the fact that (1) the direct extraction of tail samples in crude Monte Carlo is often more accurate than creating a large vector of mostly zeros, and (2) importance sampling can achieve near-perfect accuracy when its proposal distribution is well calibrated to the target's tail. Overall, this project confirms that Monte Carlo techniques are effective for estimating extreme risk measures but also highlights the importance of choosing the right approach, particularly in calibrating importance sampling—to ensure accurate and reliable estimates. Extreme risk measures, the choice of simulation

method—and particularly the proper calibration of importance sampling—is critical to obtain accurate and reliable estimates.

References

1. Hastie, T., Tibshirani, R., & Wainwright, M. (2015). *Statistical Learning with Sparsity: The Lasso and Generalizations*. Chapman & Hall/CRC.

2. Klugman, S., Panjer, H., & Willmot, G. (2012). *Loss Models: From Data to Decisions*. Wiley.
3. The basics of Value at Risk and Expected Shortfall–R-bloggers

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Appendix: R Code for Simulation and Estimation

Here you can find the code I used to frame this report, it should be running fine.

```
##########################################
# Part 1: Normal Model (N(0,1))
##########################################

# true_es
set.seed(10)
p <- 0.01
n <- 10000
true_ES_norm <- dnorm(qnorm(1-p)) / p
cat("True ES =", true_ES_norm, "\n\n")
# Crude Monte Carlo - Complete Vector Approach
Z <- rnorm(n)
threshold <- qnorm(1-p)

MCvalues <- rep(0, n)
for(i in 1:n) {
  if(Z[i] > threshold) {
    MCvalues[i] <- Z[i]
  }
}
ES_MC <- mean(MCvalues) / p
var_MC <- var(MCvalues) / n
lower_CI <- ES_MC - 1.96 * sqrt(var_MC)
upper_CI <- ES_MC + 1.96 * sqrt(var_MC)
cat("Crude MC (normal, complete vector) : ES =", ES_MC, "\n")
## Crude MC (normal, complete vector) : ES = 3.075687
cat("95% CI : [", lower_CI, ",", upper_CI, "]\n")
# Crude Monte Carlo - Direct Extraction
Y <- Z[Z > threshold]
ES_MC2 <- mean(Y)
var_MC2 <- var(Y) / n
lower_CI2 <- ES_MC2 - 1.96 * sqrt(var_MC2)
```

```r
upper_CI2 <- ES_MC2 + 1.96 * sqrt(var_MC2)
cat("Crude MC (normal) : ES =", ES_MC2, "\n")
cat("95% CI : [", lower_CI2, ",", upper_CI2, "]\n\n")
# Importance Sampling for the Normal Model
set.seed(10)
mu_IS <- 2.33
Y_IS <- rnorm(n, mean = mu_IS, sd = 1)
w <- dnorm(Y_IS) / dnorm(Y_IS, mean = mu_IS, sd = 1)
ind <- (Y_IS > threshold)
p_IS <- mean(ind * w)
num_IS <- mean(Y_IS * ind * w)
ES_IS <- num_IS / p_IS

var_IS <- var(Y_IS[ind] * w[ind]) / n
lower_CI_IS <- ES_IS - 1.96 * sqrt(var_IS)
upper_CI_IS <- ES_IS + 1.96 * sqrt(var_IS)

cat("Importance Sampling (normal) : ES =", ES_IS, "\n")
## Importance Sampling (normal) : ES = 2.66597
cat("95% CI : [", lower_CI_IS, ",", upper_CI_IS, "]\n\n")

########################################
# Part 2: Exponential Model (Exp(10))
########################################
# tru_exponential ES
set.seed(20)
lambda <- 10
VaR_exp <- -log(p) / lambda
true_ES_exp <- VaR_exp + 1 / lambda
cat("True_exponential ES =", true_ES_exp, "\n")
## True_exponential ES = 0.560517
# Crude Monte Carlo for Exp(10)
X <- rexp(n, rate = lambda)
MCvalues_exp <- rep(0, n)
for(i in 1:n) {
  if(X[i] > VaR_exp) {
    MCvalues_exp[i] <- X[i]
  }
}
ES_MC_exp <- mean(MCvalues_exp) / p
var_MC_exp <- var(MCvalues_exp) / n
lower_CI_exp <- ES_MC_exp - 1.96 * sqrt(var_MC_exp)
upper_CI_exp <- ES_MC_exp + 1.96 * sqrt(var_MC_exp)

cat("Crude MC (exponential) : ES =", ES_MC_exp, "\n")
cat("95% CI : [", lower_CI_exp, ",", upper_CI_exp, "]\n")
# Importance Sampling for Exp(10)
lambda_IS <- 10
X_IS <- rexp(n, rate = lambda_IS)
w_exp <- (lambda / lambda_IS) * exp(-(lambda - lambda_IS) * X_IS)
ind_exp <- (X_IS > VaR_exp)
p_IS_exp <- mean(ind_exp * w_exp)
num_IS_exp <- mean(X_IS * ind_exp * w_exp)
ES_IS_exp <- num_IS_exp / p_IS_exp

var_IS_exp <- var(X_IS[ind_exp] * w_exp[ind_exp]) / n
lower_CI_IS_exp <- ES_IS_exp - 1.96 * sqrt(var_IS_exp)
upper_CI_IS_exp <- ES_IS_exp + 1.96 * sqrt(var_IS_exp)

cat("Importance Sampling (exponential) : ES =", ES_IS_exp, "\n")
cat("95% CI : [", lower_CI_IS_exp, ",", upper_CI_IS_exp, "]\n")
```